Report: Professional Practice

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ShareMaq
Automated Machinery Management Firm

Abstract

ShareMaq is a Chilean technology company whose platform automates the management and administration of construction machinery. By connecting construction companies, rental providers, distributors, operators, financial services, and maintenance providers within a single ecosystem, the platform streamlines operations and drives efficiency across the industry.

I carried out my professional internship at ShareMaq as part of the engineering department, a team of five responsible for maintaining and evolving the company's two main applications. My contributions included fixing reported bugs, implementing new features, and collaborating on the redesign of the platform's core payment calculator. My work spanned from adding new fields to the Report creation form, to developing significant extensions for Workflows, one of the most important new components of the application.

Having completed my pre-professional internship at the same company, I was already familiar with the team, the codebase, and the development workflow. This allowed me to skip the typical onboarding period and contribute meaningfully from the very first day, supported throughout by my supervisor and teammates.

What set this internship apart from the previous one was the level of responsibility involved. Rather than being guided through tasks, I was expected to independently research each problem, evaluate possible approaches, and propose a solution before beginning any implementation. This degree of autonomy brought me closer to what professional software engineering actually looks like, and it represents the most valuable aspect of the experience. The internship strengthened both my technical capabilities and my ability to think and act as an engineer, leaving me significantly better prepared for the professional life ahead.

1. Introduction

1.1 Company Description

ShareMaq is a Chilean technology company co-founded by Daniela Vásquez, Vicente Bascuñán, and Matías Lyon with the goal of bringing digital transformation to the construction industry. The company began with funding from friends and family, and later obtained three grants from Corfo (Semilla, Expande, and Escala) which allowed it to develop a collaborative platform focused on automating the management and administration of construction machinery.

ShareMaq currently operates in Chile, Mexico, and Colombia, serving a client base of over 500 companies across the construction ecosystem, including equipment rental providers, distributors, financial service providers, and maintenance companies. The company's annual revenue surpasses USD$1,000,000.

Its core product is a SaaS (Software as a Service) platform that integrates various tools to simplify and streamline operational processes within the industry. Through the platform, companies can track equipment in real time, automate rental workflows, optimize machinery usage, and monitor Environmental, Social, and Governance (ESG) metrics. These capabilities translate into operational cost reductions of between 20% and 30%. An investment from View Capital has further accelerated the company's growth, enabling an expansion into automated equipment rentals and consolidating ShareMaq's position as a leading player in construction industry digitalization.

ShareMaq's organizational structure nowadays is led by CEO Matías Lyon, who oversees the company's vision and strategy. The company is divided into three main departments: Engineering, Product, and Data. The Engineering Department is headed by CTO Julio Cortés and is composed of two Software Engineers, Camila Pizarro and Luis Corvalán, and two interns, Chiara Romanini and myself. The Product Department is led by Product Owner Humberto Camelio, with Christian Cáceres as Product Business Developer and Liz Cajamarca handling Customer Service. Finally, the Data Department, which operates under the Product Department, is composed of Data Analyst Francisco Soto.

In the office I worked the organization chart looks like this:

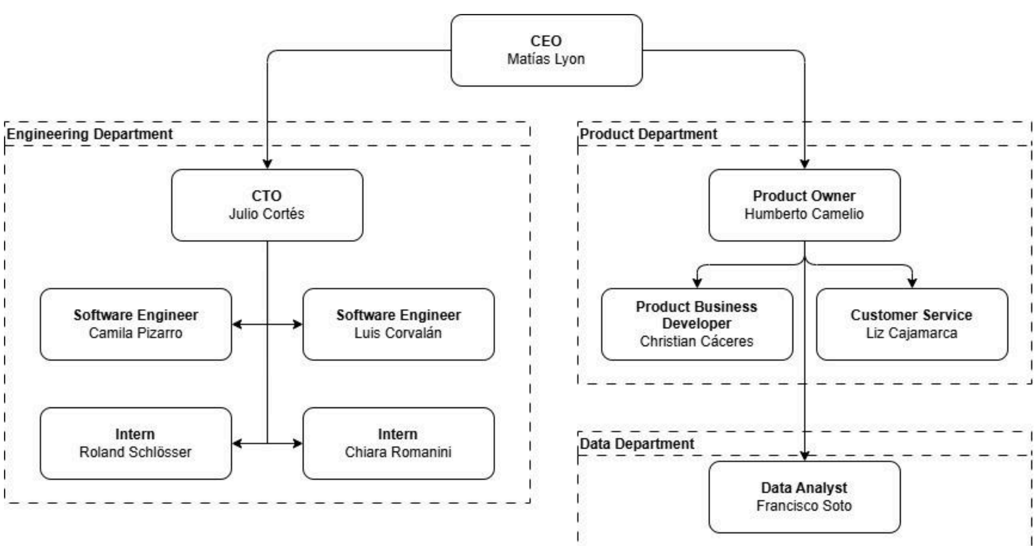

Image 1: Organizational chart of Sharemaq

As part of its international growth strategy, ShareMaq is targeting a doubling of its client base by 2026, leveraging its presence in Mexico and Colombia. Mexico represents a particularly strategic market, as its construction industry is roughly ten times the size of Chile's and could serve as a stepping stone toward entering the United States. Through ongoing innovation and continued expansion, ShareMaq aims to redefine how construction equipment is managed, driving the industry toward greater efficiency, transparency, and connectivity.

1.2 Department/Area Description

The Engineering Department at ShareMaq is made up of a team of five professionals dedicated to software development and technical implementation. The team's main responsibilities revolve around resolving bugs and developing new features requested by the Product Department, making the relationship between these two departments central to the company's day-to-day operations.

To maintain close coordination, the Engineering and Product teams hold joint meetings three times a week on Mondays, Wednesdays, and Fridays. These sessions serve as a space to discuss ongoing work, align priorities, and ensure that technical development remains consistent with the company's operational needs and product direction.

As a developer within the Engineering Department, my work spanned both of ShareMaq's main applications. My responsibilities included implementing new features, fixing reported issues, writing clean and maintainable code, and participating in the peer review process.

My direct superior was the CTO, Julio Cortés, who held a central role within the department both as its principal developer and as the team's technical lead. He facilitated our recurring meetings, defined development priorities, and provided guidance whenever the team faced technical decisions.

Our working relationship was built on open communication and mutual respect. He offered technical mentorship that helped deepen my understanding of ShareMaq's technology stack and pushed me to approach problems more rigorously. His broad knowledge of the platform's architecture and business context allowed him to steer the team's efforts effectively, and his feedback throughout my internship was instrumental in shaping the quality of my contributions.

As a team, we followed a peer review system for all code contributions. Whenever a team member completed a task, they would assign a reviewer, another developer on the team, who was responsible for examining the code before it could be approved and deployed to the production environment. This process ensured code quality and consistency across the platform, while also fostering knowledge sharing within the team.

2. Work Details

2.1 General Activities

During my two-month professional internship at ShareMaq, I worked across the full technology stack of the company's two main applications, contributing to a variety of software development tasks. Communication within the team was handled through Microsoft Teams, while task management was organized via Trello, where each bug fix or feature was tracked through its own card from assignment to completion.

Reports Development

I implemented a new field in the Report creation form, which required changes across the backend to handle the new data, as well as updates to the frontend form and its validation logic to present the field correctly to the user.

Workflows Development

Workflows is a new and critical feature of the platform that allows users to define and apply a series of processes to a construction asset. While the existing implementation only supported the creation of new assets, I extended this functionality by implementing two new workflow types: 'Edit Asset' and 'Transfer Asset', both of which operate on existing assets. This involved both backend logic to handle the new workflow types and frontend work to build the interfaces that allow users to select and interact with existing assets. I also introduced a new requirement type called Contract Condition, grouping all cost-related attributes such as rate, dates, and unit charge, with corresponding frontend fields added to the workflow form. Finally, I developed a backend service to perform a backfill of asset data into existing workflows, ensuring data consistency across the platform.

Payment Calculator Development

Towards the end of my internship, I collaborated with my teammate Camila Pizarro on migrating and refactoring the Payment Calculator, the core component responsible for generating all payment receipts. The calculator had been poorly modelled, and our work involved restructuring both the backend logic and the frontend components that depend on it, setting a cleaner foundation for future development.

Cross-Functional Collaboration

My role involved regular interaction with members across different departments. All completed work went through a peer review process, where fellow developers examined the code for quality and correctness before it could be approved for deployment. Beyond being a quality control mechanism, these reviews were also a consistent source of learning, as the feedback I received helped me better understand the team's standards and best practices.

I also maintained direct communication with the Product Department when working on bugs or features that had an impact on production workflows. Their input was essential for ensuring that the technical solutions I implemented actually addressed the underlying business needs, helping to keep development closely aligned with real user requirements.

Collaborative Planning Process

A key part of ShareMaq's development workflow was the refinement and implementation plan stage that preceded any significant task. Before beginning development, each team member was expected to research their assigned task and present a proposed technical approach to the rest of the team. This created a structured space for discussion, where different perspectives could surface potential issues early and collective knowledge could inform better solutions.

For me, these sessions were particularly valuable. Preparing and presenting an implementation plan pushed me to think through problems more thoroughly and independently, while the feedback from more experienced developers helped me refine my approach and avoid pitfalls I might not have anticipated on my own.

2.2 Most Challenging Activity

Independent Problem-Solving in a Production Environment

The most significant engineering challenge I faced during my professional internship was a fundamental shift in how I was expected to approach problems. Unlike my previous experience at ShareMaq, where I was guided more closely through tasks, this internship demanded a higher degree of autonomy. I was responsible not only for implementing solutions, but for researching the problem, understanding its scope, designing a viable approach, and presenting it to the team before writing a single line of code.

This change in responsibility level represented the biggest engineering challenge of the internship, and it manifested most clearly in two major tasks: the Workflows feature extension and the Payment Calculator migration.

The Workflows Feature

Workflows is one of the most important new components of ShareMaq's platform, allowing users to define and apply a series of processes to a construction asset. When I joined the internship, the feature was already partially developed but only supported the creation of new assets. I was tasked with extending it to support two new workflow types, editing and transferring existing assets, as well as introducing a new requirement type called Contract Condition, which encompasses all cost-related attributes such as rates, dates, and unit charges.

The challenge was not just technical. The Workflows component was relatively new, meaning there was limited internal documentation and no established patterns to follow for the extensions I had to implement. I had to study the existing code deeply to understand the design decisions behind it before I could propose anything new. This involved reading through both the backend logic and the frontend components, tracing how data flowed through the system, and understanding how the existing "Create Asset" workflow was structured so I could replicate and adapt it correctly.

Once I had a working understanding, I prepared an implementation plan that I presented to the team. This was a critical step, the feedback I received during that meeting helped me identify edge cases I had not considered, particularly around how existing asset data should be loaded and validated when a user selects an asset to edit or transfer.

A further challenge within the Workflows component was developing a backfill service, a process responsible for populating existing workflows with asset data that had not been captured at the time of their creation. This required careful design to ensure that the service could run without disrupting live data, handle edge cases where data might be incomplete, and remain maintainable for future use. The risk here was high, since a poorly executed backfill could corrupt existing records in the production database. I addressed this by designing the service to be idempotent and testable in isolation, and by validating it thoroughly in a staging environment before it was approved for deployment.

The Payment Calculator Migration

The most technically demanding task of my internship was the migration and refactoring of the Payment Calculator, which I carried out in collaboration with my teammate Camila Pizarro. The calculator is the core component responsible for generating all payment receipts within the platform, making it one of the most critical and sensitive parts of the application. The existing implementation had been built incrementally over time without a strong underlying model, resulting in a structure that was difficult to maintain, hard to extend, and prone to errors.

The first challenge was understanding the existing system well enough to redesign it. Before writing any new code, Camila and I had to map out the current logic in full, tracing every calculation, every dependency, and every edge case the calculator handled. This process revealed just how deeply embedded the calculator was in both applications, with components across the frontend and backend relying on its outputs in ways that were not always obvious.

From there, we worked together with the broader team to design a new database schema that would better support the calculation requirements. This involved several rounds of discussion and revision, as each proposed model had to be evaluated not only for correctness but for performance and compatibility with the existing data. Reaching consensus on the final design was itself a challenge, as different team members had different priorities and constraints to consider.

The migration itself was planned to be executed incrementally, moving one piece of the application at a time to avoid breaking functionality in production. However, given the scale and complexity of the task, the full migration extended beyond the internship period. By the end of my internship, Camila and I had completed the design and planning phases, laying the groundwork for the migration. This work was significant enough that I continued contributing to it after the internship concluded, transitioning into a part-time role at ShareMaq to see it through.

Key Takeaways

Both of these challenges shared a common thread: the need to think like an engineer rather than just a developer. The technical implementation, while demanding, was secondary to the process of understanding the problem, designing a sound solution, and coordinating with the team effectively. The mistakes I made were valuable precisely because they happened early enough in the process to be corrected without consequence. This internship gave me a clear picture of what professional software development looks like when the stakes are real, and it pushed me to develop the kind of independent judgment that cannot be taught in a classroom.

Implementation Planning Meetings

Preparing for these sessions required me to thoroughly investigate each task before writing any code, mapping out how a proposed feature or fix would interact with the backend, frontend, and other parts of the system. Presenting this to the team then opened the door to targeted feedback that often highlighted considerations I had not thought of on my own.

Beyond the technical value, these meetings provided a natural space to ask questions and resolve uncertainties without interrupting the team's day-to-day flow. The concentrated exchange of knowledge that took place in these discussions allowed me to progress much faster than I would have through independent exploration alone.

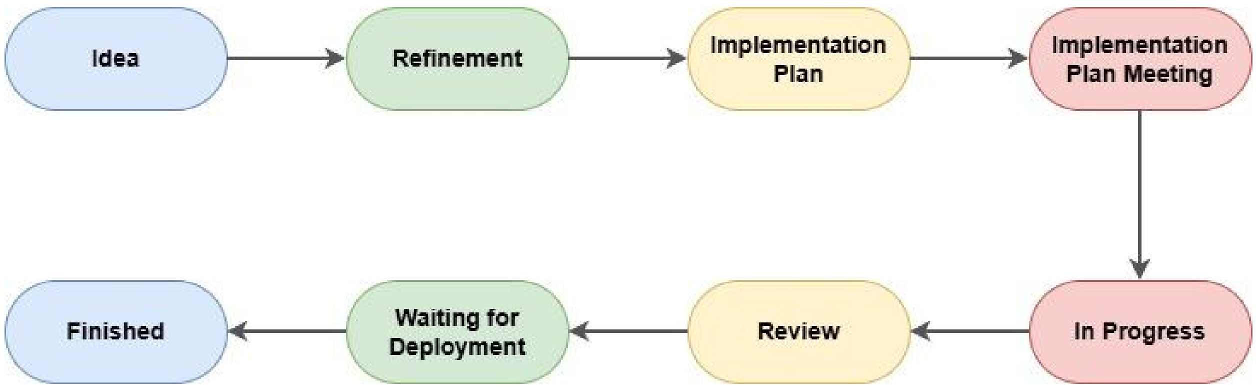

Image 2: Normal workflow of a single task from beginning to end

Quality Assurance and Error Prevention

ShareMaq's quality assurance process played an important role in ensuring that development work met the necessary standards before reaching production. The team followed a multi-stage review process designed to catch issues early and minimize risk:

- **Self-Verification:** Before submitting any work for review, each developer was responsible for thoroughly testing their own changes to confirm correct functionality and avoid introducing regressions.
- **Peer Review:** All code was then reviewed by another member of the engineering team, who assessed both the implementation quality and functional correctness. This step frequently surfaced oversights or opportunities for improvement that had been missed during self-verification.
- **Scheduled Deployment Windows:** Deployments were limited to Mondays and Thursdays, which meant that approved changes went through an additional buffer period before reaching production, allowing for any last-minute validation.
- **Hotfix Mechanism:** In the rare cases where an issue made it through to production, a streamlined hotfix process was in place to ensure a swift resolution with minimal impact on users.

This layered approach created a safety net that kept the risk of errors reaching end users low, while also providing a structured environment in which mistakes could be identified and corrected as part of the normal development cycle.

3. Analysis

3.1 Company Analysis

Organizational Structure and Effectiveness

ShareMaq's organizational structure reflects a lean and pragmatic approach suited to a company at its stage of growth. Under the strategic direction of CEO Matías Lyon, the company has built a structure that prioritizes agility and cross-departmental alignment over rigid hierarchy.

A notable example of this efficiency is the Engineering Department, which consists of just five members yet is responsible for maintaining two application platforms while continuously delivering new features. This is made possible by well-defined workflows, clear communication channels, and a close working relationship with the Product Department that keeps technical development grounded in actual business and user needs.

The reach of the Sales and Customer Success teams further reflects the organization's effectiveness. Having grown to over 500 clients across three countries, and within an industry not traditionally associated with digital adoption, demonstrates the company's ability to communicate the value of its platform to a demanding and varied client base.

Productivity and Resource Utilization

ShareMaq's productivity is reflected in several concrete indicators. The ability to maintain and continuously improve two application platforms with a small engineering team speaks to a well-organized development process and efficient use of available resources. The twice-weekly deployment schedule reinforces this, striking a practical balance between delivering new functionality quickly and ensuring that quality standards are upheld before changes reach production.

The platform's impact on clients is equally telling. By automating equipment management processes, ShareMaq enables its clients to reduce operational costs by 20 to 30%, a tangible outcome that strengthens the platform's value proposition and supports long-term client retention.

From a financial standpoint, the company has shown a consistent ability to translate investment into growth. Starting from friends and family funding, ShareMaq progressed through three Corfo grants (Semilla, Expande, and Escala) before securing investment from View Capital. Each stage of funding has corresponded with measurable expansion, reflecting a disciplined approach to resource allocation and a clear trajectory of development.

Technological Capacity and Innovation

ShareMaq's primary competitive advantage lies in the depth and breadth of its platform's capabilities. By integrating real-time equipment tracking, rental process automation, machinery management, and ESG monitoring into a single solution, the company addresses multiple pain points of the construction industry simultaneously, combining technical sophistication with a strong understanding of the sector's operational realities.

The platform's ability to support operations across Chile, Mexico, and Colombia, each with its own regulatory landscape and business practices, reflects sound architectural decisions made early in the product's development and sustained through deliberate technical planning.

The recent expansion into automated equipment rentals signals that ShareMaq operates with a mindset of continuous improvement rather than simply maintaining what already exists. Combined with the engineering team's adoption of modern development practices, this positions the company well for sustained growth and ongoing innovation as it continues to scale.

Market Position and Growth Trajectory

ShareMaq has established a solid market position, with annual revenues exceeding USD$1,000,000 across its three operating countries. Its international expansion strategy reflects a clear understanding of market opportunity, particularly in Mexico, where the construction industry is roughly ten times the size of Chile's and represents a potential entry point into the broader North American market.

The platform's ESG monitoring capabilities also position the company well in light of growing regulatory and market pressure around sustainability in the construction sector, aligning ShareMaq's offering with an industry trend that is only likely to intensify in the coming years.

Based on what I observed during my internship, ShareMaq appears to have the foundations needed to meet its growth objectives. The combination of a mature and scalable platform, efficient internal processes, and a well-defined international strategy gives the company a strong basis for continued expansion in the construction technology sector.

3.2 Work Analysis

Contribution to the Engineering Department and Organization

Throughout my professional internship, my work spanned three main areas of ShareMaq's platform: Reports, Workflows, and the Payment Calculator. Each of these contributions had a direct impact on the product's functionality and, by extension, on the experience of the company's clients.

The first task, adding an odometer field to the Report creation form, was relatively contained in scope but required coordinated changes across the backend, frontend, and validation layers. While small, it addressed a real gap in the data that clients could capture through the platform, improving the completeness of operational records for equipment management.

The bulk of my contribution was concentrated in the Workflows component, which represents one of the most strategically important features currently under development at ShareMaq. My work here can be broken down into five specific tasks: adding a selectable type option to workflow requirements, implementing "Edit Asset" workflow templates, introducing the Contract Condition requirement type for rate and cost assignment, developing the retroactive backfill service for existing Edit Asset workflows, and implementing the “Transfer Asset” workflow template. Collectively, these extensions transformed Workflows from a feature that only supported asset creation into one capable of handling the full asset lifecycle: creation, editing, and transfer. This significantly broadened the feature's utility for clients and moved it closer to being a complete, production-ready component of the platform.

The Payment Calculator work, carried out in collaboration with Camila Pizarro, addressed one of the most structurally problematic parts of the codebase. The existing calculator had been built without a solid underlying data model, which made it error-prone and difficult to extend. Our work focused on redesigning the database schema and restructuring the logic to create a more maintainable and scalable foundation. Although the full migration extended beyond the internship period, the design and planning work completed during it established the direction for all subsequent development.

Quantitative Overview

The following table summarizes the tasks completed during the internship, their scope, and the layers of the application affected:

#	Task	Scope	Layers Affected
1	Odometer field in Reports	Low	Backend, Frontend
2	Selectable type in workflow requirements	Medium	Backend, Frontend
3	Edit Asset workflow templates	High	Backend, Frontend
4	Contract Condition requirement type	High	Backend, Frontend, DB
5	Retroactive backfill service	High	Backend, DB
6	Transfer Asset workflow templates	High	Backend, Frontend
7	Payment Calculator redesign	Very High	Backend, Frontend, DB

Table 1: Tasks performed during my professional internship

Of the seven tasks completed, five were classified as high or very high complexity, meaning they required independent research, an implementation plan presentation, peer review, and validation in a staging environment before deployment. This distribution reflects the degree of autonomy and responsibility expected during the professional internship compared to the pre-professional one.

Mistakes and Corrective Measures

Two notable mistakes occurred during the internship, both of which offered concrete learning opportunities.

The first arose during the Workflows extension. In my initial implementation plan for the Edit Asset workflow, I incorrectly assumed that the data structure used for new asset creation could be reused directly when loading an existing asset. This assumption was caught during the implementation plan meeting before any code was written, which meant it had no impact on the codebase. The corrective measure here was process-based: the implementation plan stage proved its value precisely by catching this kind of conceptual error early. Going forward, a specific measure to minimize similar mistakes would be to explicitly map out the data flow for both the new and existing cases side by side during the research phase, before preparing the implementation plan.

The second mistake was related to the backfill service. During initial testing, the service was not fully idempotent, meaning that running it more than once on the same data could produce duplicate entries. This was identified during testing in the staging environment before deployment. The fix involved adding existence checks before each insert operation. As a preventive measure for future services of this kind, a useful standard would be to define idempotency as an explicit design requirement from the outset, rather than addressing it retroactively during testing.

Overall Impact

The work completed during this internship contributed directly to the growth of the Workflows feature, which is central to ShareMaq's near-term product roadmap. By extending its functionality to cover asset editing and transfer, the feature became applicable to a significantly larger share of the real-world scenarios that ShareMaq's clients encounter. The Payment Calculator redesign, while still ongoing, addresses a long-standing structural debt in the codebase that had been limiting the team's ability to extend and maintain one of the platform's most critical components. Both contributions align with ShareMaq's broader goal of scaling its platform to support a growing international client base.

4. Personal Reflection

Returning to ShareMaq for my professional internship was a unique experience. Unlike most of my peers, who were stepping into a professional environment for the first time, I already knew the team, the codebase, and the culture. What I did not fully anticipate was how different the experience would feel despite that familiarity. The tasks were harder, the expectations were higher, and the degree of guidance was significantly lower. In many ways, starting from a place of comfort made the challenge more visible. I could clearly see the gap between where I had been and where I was now expected to perform.

Professionally, this internship pushed me in ways the previous one did not. Being responsible for researching a problem, designing a solution, and defending it in front of the team before writing a single line of code forced me to think differently. It was no longer enough to understand how to implement something, I had to understand why a particular approach was the right one, and be able to justify that reasoning to experienced developers. This shift in responsibility was, without question, the most significant source of growth during the internship.

The work itself reflected that growth. The Workflows feature required me to navigate an underdocumented, actively evolving part of the codebase and extend it in multiple directions without breaking what already existed. The Payment Calculator project demanded an even higher level of thinking, involving database modelling, cross-team discussion, and long-term planning for a component that sits at the core of the platform. The fact that this work extended beyond the internship and led to a part-time position at ShareMaq is perhaps the clearest indicator of the trust the team placed in my contributions.

On a human level, this experience clarified something important for me: the kind of engineer I want to be. Working alongside teammates who approached problems with rigor, ownership, and a willingness to collaborate openly showed me that technical skill alone is not what makes someone valuable in a professional setting. The ability to communicate clearly, acknowledge uncertainty, and work through difficult problems as part of a team matters just as much, if not more.

As I finish my studies and move toward full-time professional life, I feel genuinely prepared in a way that academic work alone could not have provided. This internship did not just teach me new tools or frameworks, it gave me a realistic and demanding preview of what it means to work as an engineer, and confirmed that it is exactly what I want to be doing.